

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Security Incident Mapping (SIM)

Course: SEN202 – Software Engineering

Project Title: Security Incident Mapping (SIM)

NAME: YASERE OKIKILOLA PATRICK

Matric Number: 241201208

Department: COMPUTER SCIENCE

Table of Contents

1. Introduction
2. Purpose
3. Scope
4. Objectives
5. Stakeholders
6. System Description
7. Functional Requirements
8. Non-Functional Requirements
9. System Modules
10. Assumptions and Constraints
11. Future Improvements
12. Conclusion

1. Introduction

Security incidents such as theft, assault, rape, vandalism, and other crimes have become common on many university campuses. Victims often find it difficult to report incidents promptly, making it difficult for security personnel to respond quickly.

Security Incident Mapping (SIM) is a web-based application designed to provide students, staff, and landlords with a simple platform to report security incidents occurring within and around the campus. The system records incident information, location, description, and category to enable security officers to respond effectively.

2. Purpose

The purpose of this project is to develop a simple web application that enables members of the campus community to report security incidents electronically while helping security personnel monitor reported cases.

3. Scope

The SIM system allows users to:

- Register/Login
- Report security incidents
- Select incident location
- View reported incidents
- Contact emergency services
- Allow administrators to manage reports

The system is intended for campus use only.

4. Objectives

The objectives of the system include:

- Reduce campus crime.
- Encourage quick reporting of incidents.
- Improve communication between students and security officers.
- Maintain a digital database of reported incidents.
- Assist management in identifying crime-prone areas.

5. Stakeholders

The stakeholders include:

- Students
- Staff
- Landlords
- Campus Security Personnel
- School Management
- System Administrator

6. System Description

SIM is a web application developed using HTML, CSS and JavaScript.

Users submit incident reports through an online form.

Each report contains:

- Incident Type
- Date
- Time
- Location
- Description
- Reporter Information

The administrator can review submitted reports and monitor security trends.

7. Functional Requirements

The system shall:

1. Allow users to register.
2. Allow users to login.
3. Allow users to report security incidents.
4. Allow users to upload evidence (optional).
5. Store incident records.
6. Display reported incidents.
7. Allow administrators to delete false reports.
8. Provide emergency contact information.
9. Allow searching of incidents.
10. Generate simple statistics of reported incidents.

8. Non-Functional Requirements

The system should:

- Be easy to use.
- Load quickly.

- Be secure.
- Be available 24 hours.
- Protect users' personal information.
- Support modern web browsers.
- Have a responsive design for mobile devices.

9. System Modules

Module 1 – User Registration

Users create accounts before accessing the system.

Module 2 – Login

Registered users authenticate with username and password.

Module 3 – Incident Reporting

Users fill in a form containing:

- Incident Type
- Date
- Time
- Location
- Description

Module 4 – Incident Management

Administrators review and manage reports.

Module 5 – Emergency Contacts

Users can quickly access campus emergency numbers.

Module 6 – Dashboard

Displays summaries of reported incidents.

10. Assumptions and Constraints

Assumptions

- Users have internet access.
- Users provide accurate information.
- Security officers monitor reports regularly.

Constraints

- Internet connection is required.
- Users may submit false reports.
- The mini version does not include GPS tracking or real-time notifications.

11. Future Improvements

Future versions may include:

- Live GPS tracking
- Anonymous reporting
- SMS alerts
- Email notifications
- AI crime prediction
- Mobile application
- Interactive crime heat maps

12. Conclusion

The Security Incident Mapping (SIM) system provides a simple and efficient solution for reporting and monitoring security incidents within a university community. The application promotes faster communication between students and security personnel, thereby contributing to a safer campus environment.